

LEGACY PROTOCOL (LGY)

Technical Whitepaper — Revision 2026.3

Supersedes Rev. 2026.2 (published 2026-06). This revision corrects a governance-threshold error, documents four features added to the contract since the last revision, corrects the description of post-claim content deletion to reflect that it is scoped per-recipient rather than vault-wide, and adds full disclosure of the one known pre-mainnet cleanup item. The status-banner methodology from Rev. 2026.2 is unchanged.

A NOTE ON HOW TO READ THIS DOCUMENT

Every section below opens with a status banner showing what is actually live in the audited smart contract today, versus what remains a planned, future capability. This revision exists specifically so that investors, partners, and the public can evaluate Legacy Protocol based on what exists right now, with a clear and honest view of the build that remains.

WHAT CHANGED SINCE REV. 2026.2

- Governance threshold corrected throughout this document: the live contract uses 3-of-6 core keys, not 3-of-5 as earlier drafts stated (§3.5, §4.1, §4.4).
- New: Service Provider Registry (§3.2) — governance can now register multiple storage providers; vault creators pick one per vault.
- New: optional permanent veto waiver at vault creation (§3.4).
- New: post-claim content deletion, scoped per-recipient (§3.6) — a recipient can permanently delete their own access to a vault's content after claiming it, without affecting any other recipient of the same vault.
- New section: BTC Holder Airdrop mechanics (§4.7) — the soulbound-token airdrop contract, built and tested but not yet launched.
- Security Posture (§6) now discloses the one remaining known pre-mainnet cleanup item and the deliberate absence of any pause/guardian mechanism.

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1. Abstract & Problem Statement

In the contemporary digital landscape, information custody and digital asset transmission remain exposed to central counterparty failure. Whistleblowers, journalists, executives, and crypto holders face real obstacles when trying to preserve or transfer sensitive material, legal testimony, or private wealth in the event they are silenced, incapacitated, or pass away.

This whitepaper describes both the long-term vision for Legacy Protocol and, distinctly, the subset of that vision that exists as working, reviewed code today. The problems below are the motivation for the project; the sections that follow make clear which solutions to them are real and which are still ahead of us.

1.1 The Core Problems We Are Solving For

- **The Whistleblower Vulnerability:** informants and journalists lack a trustless method to secure “insurance vaults.” If compromised, their material is often permanently lost.
- **Single-Beneficiary Bottleneck:** most digital-asset succession tools support only one static recipient, preventing simultaneous, multi-party release of sensitive material.
- **Metadata Harvesting:** centralized discovery portals can let an attacker query a target identifier and learn whether that target has anything at stake — turning the recipient into a surveillance target before any document is ever released.

2. The Solution: Legacy Protocol (LGY)

STATUS: PARTIALLY IMPLEMENTED

Legacy Protocol is a non-custodial smart contract system deployed on the BASE network (Coinbase L2), built around the idea of a multi-tenant “Dead Man’s Switch”: a user locks an encrypted reference to their material in a vault, names one or more recipients, and the vault becomes claimable either after a chosen time period elapses or after a recipient-initiated emergency claim survives a creator-controlled challenge window.

This is the layer that exists and has been security-reviewed today. The protocol's longer-term vision — described throughout the rest of this document — layers additional infrastructure (decentralized storage, a token economy, an off-chain claims-adjudication network) on top of this working core. Each later section states plainly which parts of that additional layer are built and which remain on the roadmap.

3. Architecture

3.1 Multi-Vault IDs & Multi-Recipient Vaults

STATUS: LIVE IN CONTRACT TODAY

Each vault is created with a unique ID and can name multiple beneficiaries simultaneously. A user can designate several recipients — for example, several news outlets or several family members — who can each independently claim the same vault once it unlocks.

What this means today: Live in the audited contract today, capped at 50 recipients per vault for gas-cost safety. Earlier drafts of this document described an “unbounded” recipient array — that has been corrected here to reflect the actual, tested limit.

3.2 Decentralized Storage, Shamir's Secret Sharing & the Service Provider Registry

STATUS: PARTIALLY IMPLEMENTED

The long-term architecture calls for sharding the decryption key across a 20-of-50 Shamir's Secret Sharing scheme distributed across independent, “blind” storage nodes that have zero contextual awareness of a vault's contents or its beneficiaries.

What this means today: The node network, node staking, and client-side secret sharing described above are still not built — that part of the vision is unchanged from Rev. 2026.2, and no actual storage provider or node contract has been stood up or connected yet. What has been added to the main contract's code since that revision is a Service

Provider Registry: a governance-gated mechanism (via the same 3-of-6 multisig process described in §3.5) to register and deactivate multiple named storage-provider entries, each with its own endpoint address, so a vault creator can choose which registered provider to use at creation time (or accept the default router) instead of the contract pointing at one fixed address. This registry logic is written and compiles as part of the audited contract, but — unlike the vault lifecycle functions in §3.1, §3.4, §3.5, and §3.6 — it is not yet exercised by the current automated test suite, and no real service provider has been registered or used in practice. It should be read as implemented scaffolding for future storage flexibility, not as a battle-tested or currently operating feature. Today's vaults still simply reference an encrypted data URI; the underlying storage can be any system the creator chooses (including existing decentralized storage such as IPFS) until real storage providers are registered and the dedicated node network is built.

3.3 Private Discovery

STATUS: PARTIALLY IMPLEMENTED

To prevent attackers from harvesting vault metadata, the goal is for the public-facing application to reveal nothing about which addresses have active vaults, with matches delivered privately to the correct party only.

What this means today: The contract does not expose a public function that lists vaults by recipient unless the caller already holds the correct cryptographic credential for that vault — there is no public “search by email” feature, and no leaderboard of active vaults. However, this is what we term Private Discovery rather than the stronger “Zero-Knowledge Discovery” named in earlier drafts: a sufficiently sophisticated party with direct access to contract storage (rather than the public app) could, in principle, observe vault data. True zero-knowledge discovery requires a dedicated cryptographic proof system, which is a research-stage roadmap item (see §8).

3.4 Customizable Anti-Fraud Challenge Window

STATUS: LIVE IN CONTRACT TODAY

When an emergency claim is filed against a vault, the protocol does not release it immediately. Instead, it opens a challenge window — customizable by the vault's creator between 72 hours and 30 days — during which the creator can prove they are still active and veto the claim.

What this means today: Live today. Only an address the creator explicitly designated as a recipient (verified via cryptographic signature, not a publicly readable value) can open a claim, preventing spam claims from unrelated parties. The creator can call a veto function at any time during the window to cancel a fraudulent or mistaken claim. New since Rev. 2026.2: a creator may optionally and permanently waive their own veto right at vault creation. This trades away the safety net of a contestable claim in exchange for a guarantee to recipients that, once the challenge window opens, nothing the creator does afterward can block release — useful for creators who want to make an unconditional commitment to their recipients up front. This choice is permanent and is recorded on-chain per vault; it cannot be reversed after creation.

3.5 Governance Timelock & Multisig

STATUS: LIVE IN CONTRACT TODAY

Changes to sensitive protocol parameters — such as where data is routed, or the base pricing rate — should require both a waiting period and agreement among multiple independent parties, not a single administrator key.

What this means today: Live today: any change to the storage router, the base pricing rate, the service provider registry, or the core-key set itself requires approval from at least 3 of 6 designated core keys, plus a mandatory 14-day on-chain waiting period after the third approval before the change can take effect. No single key, including the contract deployer, can act alone or bypass the wait. Correction from Rev. 2026.2: earlier drafts of this document described a 3-of-5 threshold. The live contract uses 3-of-6. The core-key count was raised from 5 to 6 (the approval threshold stayed at 3) specifically so that if 3 keys are ever lost or compromised — through attack or simple real-world disruption, not necessarily malice — the honest remainder (3 of 6) can still reach quorum to recover or contest, where a 5-key system would have left only 2 honest keys, short of quorum, with no path to recover. This does not change the cost of an attack, which still requires obtaining the same 3 keys either way — it only improves the odds of recovery after a partial key loss.

By design, there is no pause or guardian mechanism anywhere in this contract. No single key can freeze vault creation, claims, or governance for any reason. The 3-of-6 approval requirement, the unconditional 14-day delay, and an independent 3-of-6 cancellation vote on any pending governance proposal together provide the same practical protection a pause would, without creating a single point of centralized control.

3.6 Post-Claim Content Deletion

STATUS: LIVE IN CONTRACT TODAY — NEW SINCE REV. 2026.2

Once a recipient has successfully claimed a vault, they may choose to permanently delete their own access to its encrypted content reference, using the same cryptographic-signature proof required to claim in the first place.

What this means today: Live today. Deletion requires that the vault already be claimed, and can only be triggered by an address the creator designated as a recipient, verified via the same cryptographic-signature proof used to claim. Deletion is scoped to the calling recipient only: on a multi-recipient vault, one recipient deleting their own access does not affect any other recipient's ability to independently claim the same underlying content. This is irreversible for the recipient who deletes it: once done, not even that same recipient can retrieve the reference again through this contract. This exists for recipients who want positive confirmation that they, personally, are finished with sensitive material, without forcing every other named recipient of the same vault to lose access at the same time.

4. Token Economics (Planned Model)

STATUS: PLANNED — NOT YET BUILT

This entire section describes the planned LGY token economy. None of it exists on-chain yet: there is no LGY token contract deployed, and all fees in the current, live smart contract are denominated and paid in native ETH. This section is presented as a roadmap and a basis for discussion with investors and partners, not as a description of a live system.

4.1 Supply & Distribution

STATUS: PLANNED — NOT YET BUILT

Planned fixed supply of 1,000,000,000 LGY, with protocol fees split across node operators, a growth treasury, the founding team, and a deflationary burn.

What this means today: No LGY token contract exists yet. The fee-splitting logic itself (an 80% / 8% / 7% / 5% split across node operators, marketing, the founding team, and a permanent burn address) is already implemented and tested in the live contract — it currently distributes ETH fees, not LGY. Once an LGY ERC-20 contract is written, audited, and deployed, this same distribution logic can be pointed at LGY token flows.

4.2 Activation Fee, Time-Decay Fee & Storage Surcharge

STATUS: PARTIALLY IMPLEMENTED

A flat activation fee at vault creation, a continuous time-based decay fee while a vault is active, and a one-time surcharge for larger encrypted payloads, with pricing designed to stay stable against market volatility.

What this means today: The activation fee and time-decay fee both exist and are live today, denominated in ETH at a fixed rate set by protocol governance (not yet by a price oracle — see §4.4). A vault's decay rate is locked at the rate in effect when it was created, so a later governance rate change never retroactively speeds up or slows down an existing vault's decay. The specific volumetric storage surcharge formula described in earlier drafts is not yet implemented as a separate on-chain calculation.

4.3 365-Day Post-Claim Recovery Window

STATUS: LIVE IN CONTRACT TODAY

Once a beneficiary successfully claims a vault for the first time, they should be able to re-download the material for free for a full year, protecting against local hardware failure.

What this means today: Live today. The contract enforces a 365-day window after first claim during which the same verified recipient can re-claim access at no additional cost, using a freshly signed, single-use signature for each re-download.

4.4 Oracle-Pegged Dynamic Pricing

STATUS: PLANNED — NOT YET BUILT

Long-term, fees should track real-time market conditions via a price oracle, insulating users from cryptocurrency volatility.

What this means today: Not implemented, and intentionally so at this stage: LGY has not launched and has no market price for an oracle to reference. Current pricing is a fixed ETH rate, adjustable only through the same 3-of-6 multisig and 14-day timelock process described in §3.5 (corrected from 3-of-5 in earlier drafts) — a deliberately infrequent, manual process, not a live feed. Once LGY has real market liquidity, pricing can be migrated to a live price feed (e.g. Chainlink) via a separate, purpose-built mechanism with no governance vote in its critical path; that mechanism does not exist in the contract today and is not simply “the same governance path” as other parameter changes.

4.5 Validator/Node Collateral & Slashing

STATUS: PLANNED — NOT YET BUILT

Operators of storage/routing nodes would be required to lock LGY collateral (a higher tier for whistleblower-grade nodes), with misbehavior resulting in collateral being burned.

What this means today: No node registry, staking, or slashing contract exists yet. This depends on both the LGY token (§4.1) and the decentralized storage network (§3.2) being built first.

4.6 Pre-Seed SAFT & Vesting

STATUS: PLANNED — NOT YET BUILT

Pre-seed capital is intended to be raised via a standard SAFT structure with a 12-month cliff and 24-month linear vesting.

What this means today: This is a legal and financial structure, not smart-contract logic that exists today. A vesting contract is straightforward to deploy once the LGY token contract exists, using well-established, auditable patterns; the SAFT terms themselves are a legal agreement between the company and investors, independent of the contract code.

4.7 BTC Holder Airdrop (Soulbound LGY)

STATUS: PARTIALLY IMPLEMENTED — NEW SECTION

Separately from the main vault contract, a dedicated contract governs a planned one-time airdrop of LGY to Bitcoin holders, intended to bring the BTC-holder community into the protocol at launch.

What this means today: The airdrop contract has been written and is covered by 27 passing automated tests, but it cannot go live until the LGY token itself exists, and no snapshot, registration window, or claim window has been opened yet. Its designed lifecycle is five sequential phases: (1) the owner locks in a specific Bitcoin block height as a snapshot reference; (2) a 90-day registration window opens, during which eligible BTC holders prove ownership off-chain and register a receiving ETH address; (3) once registration closes, the owner assigns each registered address its LGY allocation; (4) a 90-day claim window opens, during which users claim their allocation as Soulbound LGY — tokens that cannot be transferred to any other address and can only be spent by interacting with the main Legacy Protocol vault contract (for example, to fund a vault); and (5) any allocation left unclaimed after the claim window, or any soulbound balance not spent within 90 days of being claimed, expires and is burned. This is a planned launch mechanism with real, tested code behind it, not a live or currently running airdrop.

5. Claims Adjudication (Planned Model)

STATUS: PLANNED — NOT YET BUILT

This section describes the planned multi-stage process for verifying emergency claims (e.g. death certificates) at scale. None of the adjudication logic below exists on-chain today. What the live contract does already support is the final step common to all three planned stages: once a claim is determined valid by whatever process is used, a designated recipient can open and, if uncontested through the challenge window, complete the claim on-chain using a verified cryptographic signature.

5.1 Stage One: Automated Document Review

STATUS: PLANNED — NOT YET BUILT

Digital death certificates or registry QR codes would be parsed by an automated document-reading process that authenticates against sovereign registries directly.

What this means today: Not implemented. This is an off-chain document-processing service that does not exist yet. Once built, it would interface with the live contract only at the final step — confirming a claim that the contract already knows how to process.

5.2 Stage Two: Human Review Team

STATUS: PLANNED — NOT YET BUILT

Complex or low-confidence certificates would be routed to a small team of human reviewers who verify the registry record and post the result for peer confirmation.

What this means today: Not implemented. An off-chain workflow and review tooling would need to be built; the on-chain footprint would again be limited to confirming the final outcome.

5.3 Stage Three: Decentralized Jury

STATUS: PLANNED — NOT YET BUILT

At high transaction volume, a decentralized, collateral-backed jury pool would vote anonymously on contested claims, with dishonest jurors losing their stake.

What this means today: Not implemented. This is a substantial, standalone subsystem (collateral, anonymous voting, slashing) comparable in scope to existing decentralized-court protocols. It is sequenced as a later-phase roadmap item, not a near-term deliverable.

6. Security Posture

STATUS: LIVE IN CONTRACT TODAY

The live smart contract has been through an internal security review pass covering, among other things, reentrancy protection, front-running resistance on claim authorization (using cryptographic signatures rather than exposing secrets in public transactions), denial-of-service resistance on unbounded loops, and a real on-chain multisig-plus-timelock for sensitive parameter changes. The main vault contract's core lifecycle (creation, extension, veto, unstaking, and claimability notification) and the separate BTC airdrop contract are covered by 36 automated tests (Foundry) in total, all currently passing (9 for the main vault contract, 27 for the BTC airdrop contract). The repository's full Foundry suite reports 38 tests passing; the other 2 are the default Foundry template counter tests, unrelated to either contract, and are not part of this coverage claim.

Not yet covered by an automated test: the claim and content-deletion signature flows (§3.1/§3.6), the governance proposal/approve/execute/cancel lifecycle (§3.5), and the Service Provider Registry (§3.2) — these exist in the deployed contract code but have only been exercised manually, not by the test suite. This includes the per-recipient scoping of post-claim content deletion described in §3.6. Expanding automated coverage to these paths is planned before the third-party audit.

This internal review does not substitute for an independent third-party audit. A formal audit (e.g. via a firm such as CertiK or Hacken) is budgeted into the planned use of pre-seed funds and is treated as a prerequisite — not an optional step — before any user funds interact with the contract on a public mainnet.

Known pre-mainnet cleanup item (disclosed here for completeness): the current contract still contains a temporary debug helper function, added only to troubleshoot a signature-verification issue during manual testing. It does not

expose any secret and cannot itself be used to bypass any check, but it is test scaffolding, not intended production surface area, and is tracked as a required removal item before mainnet deployment.

No pause or guardian mechanism exists anywhere in this contract, by design. This was reached only after three separate rounds of review specifically probing for centralization risk: each time a pause-like mechanism was proposed, it was found to concentrate meaningful power in a single key without adding protection beyond what the 3-of-6 governance threshold, the unconditional 14-day timelock, and an independent cancellation vote already provide. No single key — including the original deployer — can halt vault creation, block a legitimate claim, or freeze governance for any reason.

7. Core Team

- Founder & CEO: Tomas Petrauskas — corporate operations, regulatory strategy, and venture execution.
- Co-Founder & CTO: AI Systems Partner — smart contract architecture and protocol engineering.

8. Roadmap

Sequenced by technical dependency rather than by calendar ambition — each phase requires the one(s) before it.

Phase	Scope	Depends On
Phase 0	Contract hardening: internal review complete (this document); independent third-party audit and formal compiler verification still pending.	Completed internal review
Phase 1	Public testnet launch of the core vault product (create, extend, claim, veto, cancel) with ETH-denominated fees. Contract code complete, 38/38 Foundry tests passing; Base Sepolia testnet deployment itself still pending.	Phase 0
Phase 2	LGY ERC-20 token contract, vesting contract for SAFT terms, migration of fee payments from ETH to LGY, and activation of the BTC Holder Airdrop (§4.7).	Phase 1 live and stable
Phase 3	Oracle-based dynamic pricing once LGY has real market liquidity.	Phase 2
Phase 4	Decentralized storage network and/or integration, client-side Shamir's Secret Sharing, node staking and collateral. The Service Provider Registry (§3.2) is a live, interim step toward this phase.	Phase 2 (token economics)
Phase 5	Claims adjudication system: automated document review, human review team, decentralized jury.	Phase 1 (claims flow proven in production)
Phase 6	True zero-knowledge discovery (cryptographic membership proofs) replacing today's Private Discovery model.	Research-stage; no fixed dependency

A detailed, line-by-line cross-reference of every claim in this document against the audited contract is available as a separate companion document (Whitepaper vs. Code Gap Analysis) for technical due diligence.